

UNIVERSITÀ
DEGLI STUDI
DEL
SANNIO
Benevento

Università degli studi del Sannio
Dipartimento di Ingegneria
Corso di laurea Magistrale in Ingegneria Informatica

**Un'architettura LLM-based per l'automazione delle azioni
di Personaggi Non Giocanti in videogiochi**

Relatore

Ch.mo Prof.
Massimiliano Di Penta

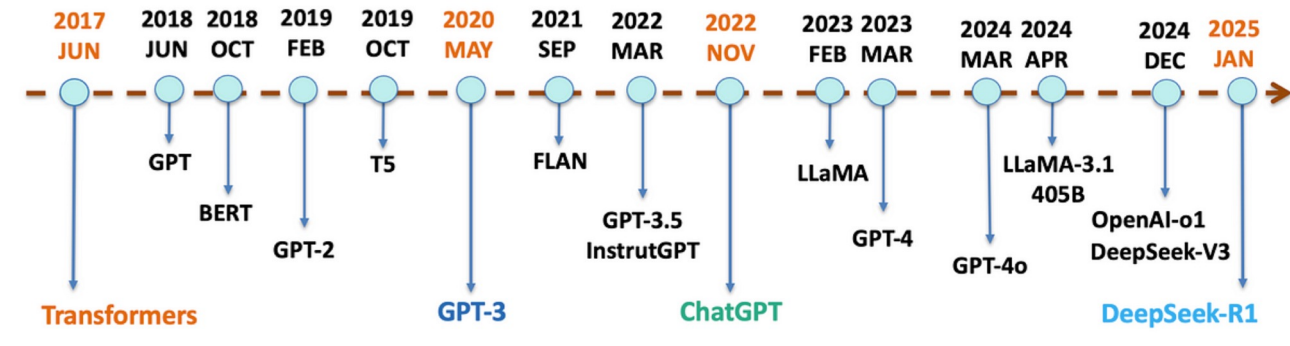
Candidato

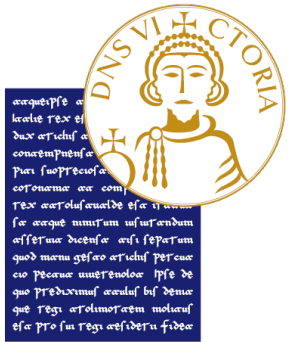
Nicola De Lillo
Matr. 399000593



IA e Videogiochi

- Negli ultimi anni, l'IA è cambiata radicalmente grazie ai Large Language Models.
- Intanto, il videogioco è diventato un mezzo culturale e artistico sempre più influente.

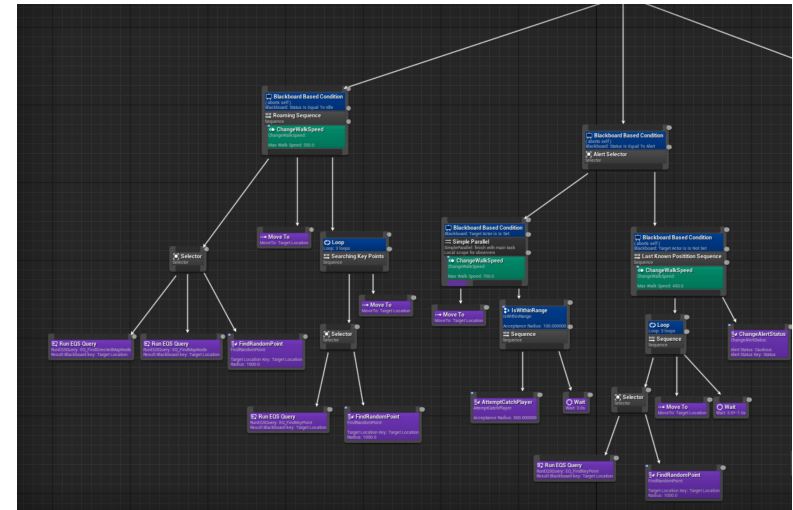




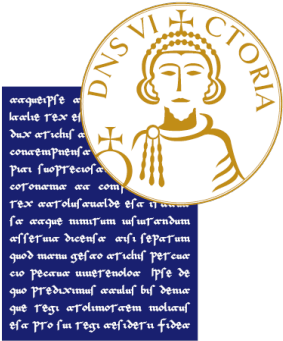
Personaggi Non Giocanti (NPC) Statici in Mondi Sempre Più Realistici



Avanzata fedeltà visiva dei mondi di gioco moderni



Architetture comportamentali tradizionali degli NPC



La ricerca attuale

Due approcci nella ricerca IA + videogiochi:

- LLM come attore decisionale
- Utilizzo degli LLM con approcci ibridi

Enhancing Game AI Behaviors with Large Language Models and Agentic AI

Ciprian Paduraru
Gameloft &
University of Bucharest
Romania

Miruna Paduraru
Electronic Arts
Romania

Alin Stefanescu
University of Bucharest &
Institute for Logic and Data Science
Romania

Abstract

Integrating advanced AI behaviors is central to creating immersive and dynamic video game experiences. This paper presents a novel approach to improving AI behaviors in games using Large Language Models (LLMs) and agent-based AI. By orchestrating various interconnected parts, we propose a framework that facilitates the creation of complex behavior trees (BTs) for non-player characters (NPCs). Our method bridges the gap between source code and visual tools in game engines and enables both technical and non-technical stakeholders to effectively contribute to the development process. We also aim to increase the diversity of observable behaviors and testability of games through the same methods. The proposed architecture is designed to be adaptable to different game engines to ensure scalability and flexibility. In a collaboration between industry and academia, we validate our approach and demonstrate its potential to improve game AI development and make it more accessible and efficient. To promote the adoption of the methods, we consider small-sized models that run on typical developer platforms without the need for external solutions or expensive computing resources.

CCS Concepts

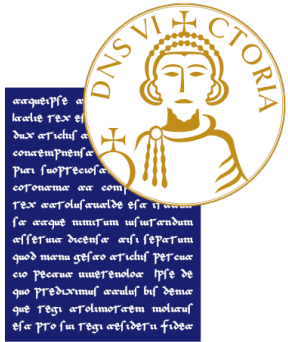
• **Human-centered computing** → *Human computer interaction (HCI)*; **Interaction techniques**; • **Computing methodologies** → **Knowledge representation and reasoning**.

1 Introduction

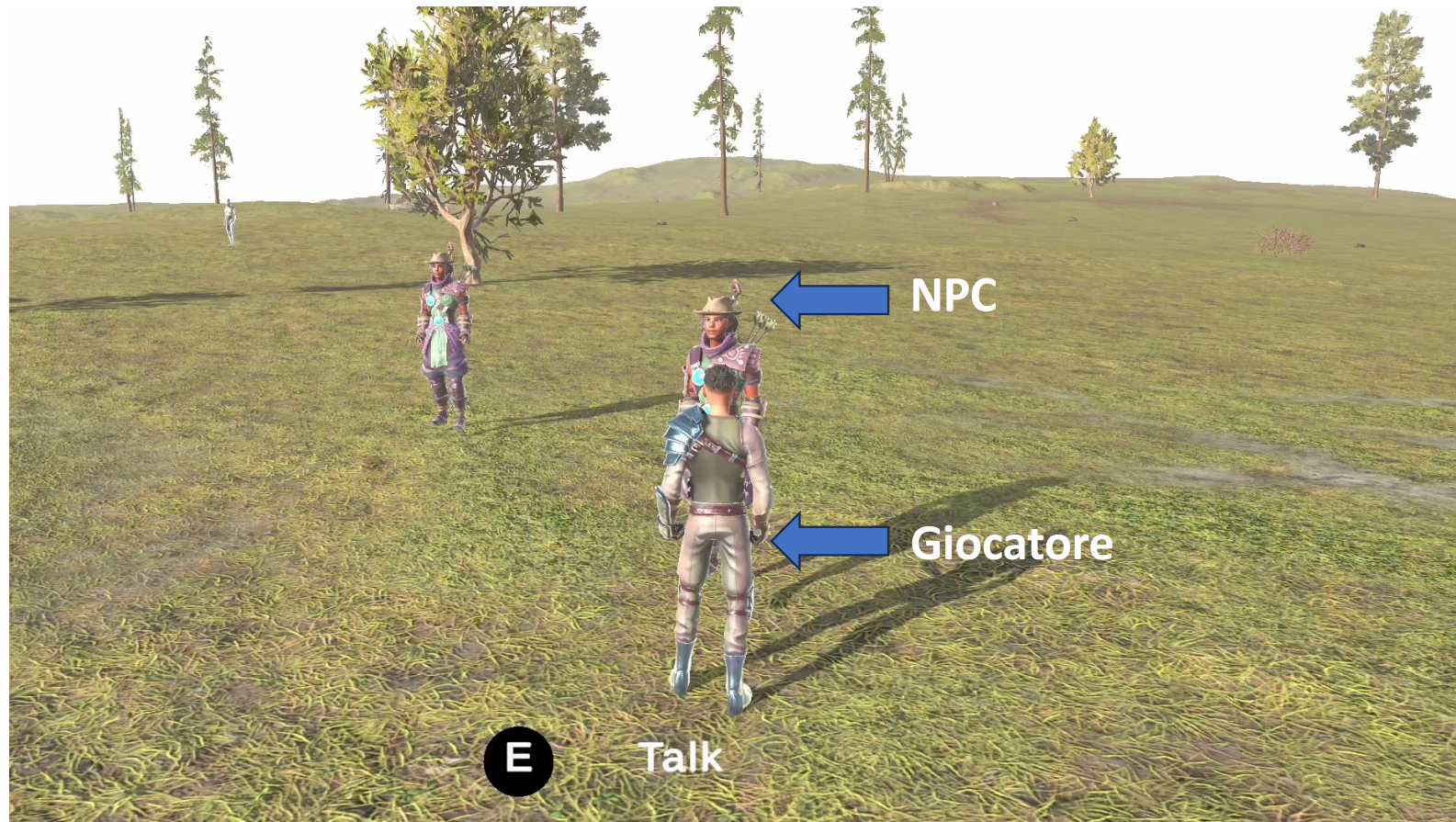
In the traditional approach to game development, programmers often write the game mechanics and AI behavior directly into the source code. This practice can be a significant barrier to active participation in the development process for those without programming skills. This limits the scalability of production and limits the ability of player communities to contribute their ideas [6].

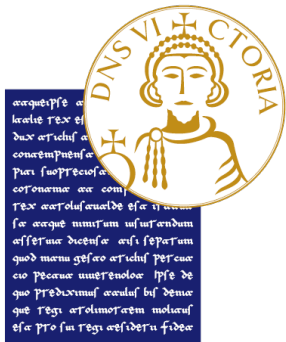
Recent advances in game engines have introduced visual tools for building complex AI systems in games [28] and simulation applications, such as behavior trees (BTs), finite state machines (FSMs), and goal-oriented action planning (GOAP). These tools aim to find a middle ground that makes it easier for a variety of users to engage in game development. However, requirements from game companies such as Electronic Arts¹, Gameloft², and Amber³ reveal that integration between the source code base and visual tools remains insufficient, leading to a divide between technical and non-technical stakeholders. The game development industry is facing an increasing demand for AI systems that provide realistic and varied NPC behaviors. However, traditional development practices often require extensive programming knowledge, creating a bottleneck for non-technical stakeholders. Additionally, integrating generative AI methods into existing visual tools and source code is complex and resource-intensive. These barriers hinder scalability and reduce the creative contributions of different stakeholders.

In a close collaboration between game development companies and academia, we worked to better understand the problem and



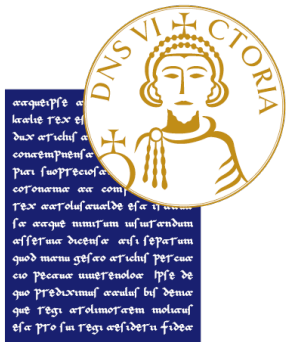
Obiettivo della tesi





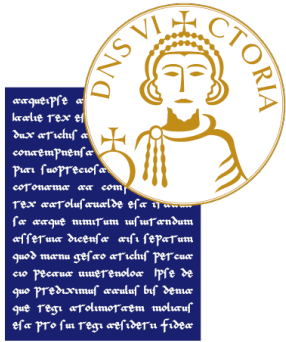
Approccio: Text-to-action

Azione	Esempio di Risposta (NPC)
 Talk	"My name is Elira."
 Flee	"I must retreat!"
 Warn Others	"Everybody, watch out!"
 Ignore	"I can't share this."
 Give Item	"Take this potion."



Approccio: Action-to-action

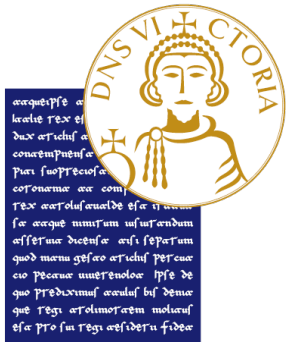
Azione	Esempio di Risposta (NPC)
👋 Greet	“Greetings, adventurer.”
👣 Approach	“Let me come closer.”



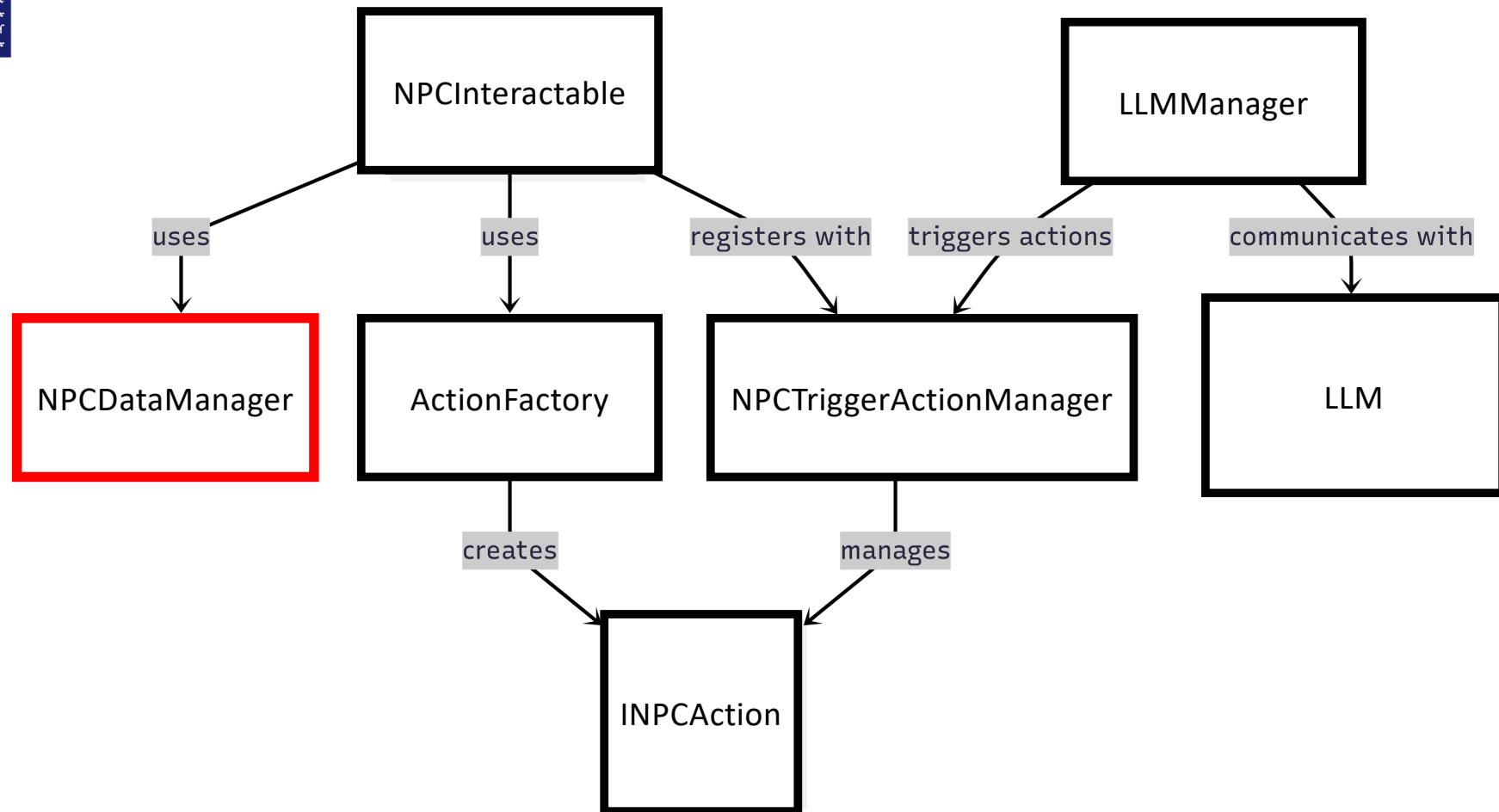
Obiettivi di Design

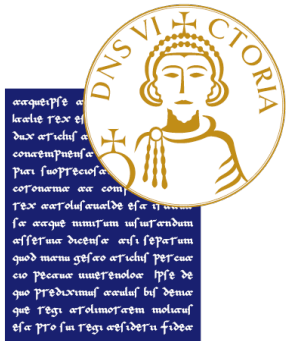
La progettazione dell'architettura è stata guidata da una serie di obiettivi chiave pensati per garantire flessibilità, estendibilità e integrabilità in un contesto videoludico real-time:

- **Separazione tra contenuto e logica**
- **Estensibilità**
- **Gestione dinamica**
- **Reazioni a catena**



Architettura del sistema





Configurazione degli NPC

Oggetti

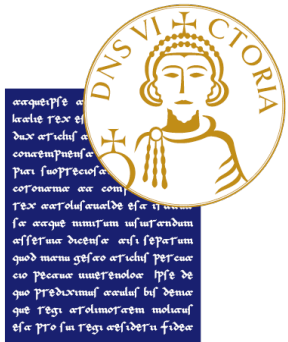


```
<GiveableItems>
<ConsumableItem>
  <tag>Potion01</tag>
  <name>Minor Potion</name>
  <description>A simple red elixir once refined by Wesely Snipes,
a famed battlefield medic of the Emerald Order. Restores 20 HP</description>
  <count>2</count>
  <modifiers>
    <StatModifier>
      <targetStat>Hp</targetStat>
      <value>20</value>
      <modifierType>Additive</modifierType>
    </StatModifier>
  </modifiers>
  <spritePath>Sprite/RPG_inventory_icons/MinorHealthPotion</spritePath>
</ConsumableItem>
```

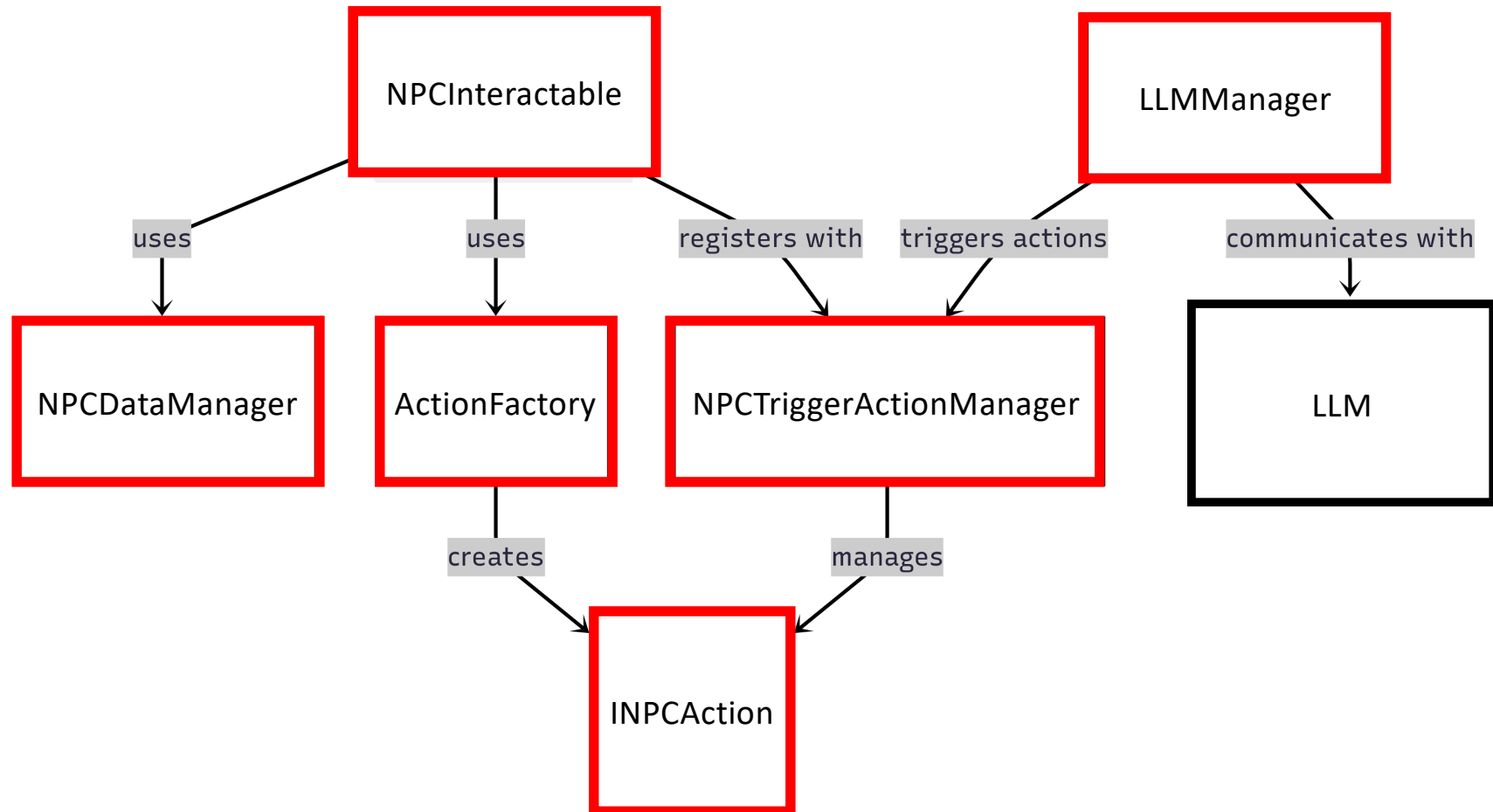
Comportamenti

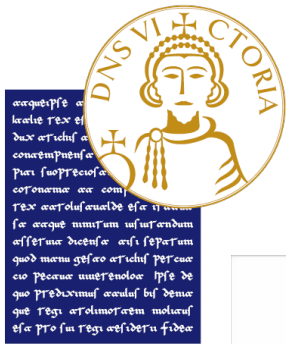


```
</GiveableItems>
<TriggerActions>
  <TriggerAction>
    <Trigger>WarnOthers</Trigger>
    <Action type="WarnOthers">
      <Parameter key="alertRadius" value="10"/>
    </Action>
  </TriggerAction>
```



Architettura del sistema





GiveItem Action





Flee Action





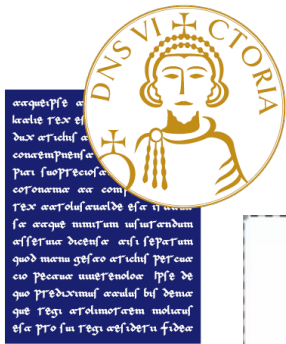
WarnOthers Action





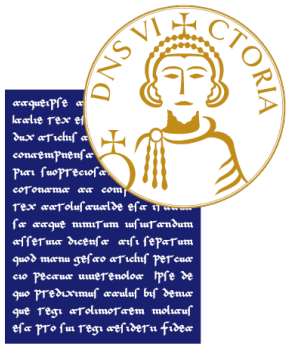
Approach Action





Greet Action

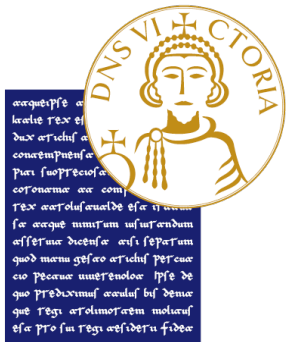




Validazione

È stato condotto uno studio su un piccolo gruppo di partecipanti considerando 4 metriche:

- Responsività
- Appropriatezza dell'azione
- Memoria e Contesto
- Coerenza

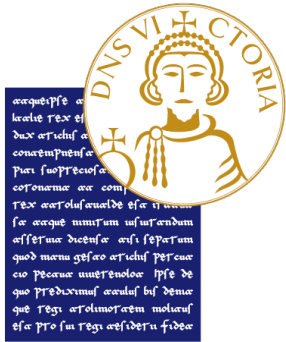


Conclusioni

Questa tesi ha affrontato il problema dell'implementazione di un'architettura LLM-based per l'automazione delle azioni dei personaggi non giocanti (NPC) in videogiochi.

La tesi ha quindi esplorato due problemi principali:

- Progettare un'architettura sostenibile e implementabile in motori di gioco reali come Unity
- Bilanciare l'autonomia decisionale del modello con la necessità di controllo e prevedibilità richiesta dai sistemi di gioco.



Limitazioni e Sviluppi futuri

Nonostante i risultati positivi ottenuti, l'implementazione ha evidenziato diverse sfide tecniche che definiscono le direzioni più promettenti per la ricerca futura:

- La latenza di risposta
- Integrazione multimodale
- Bilanciare l'autonomia dell'IA con il controllo creativo del game designer.



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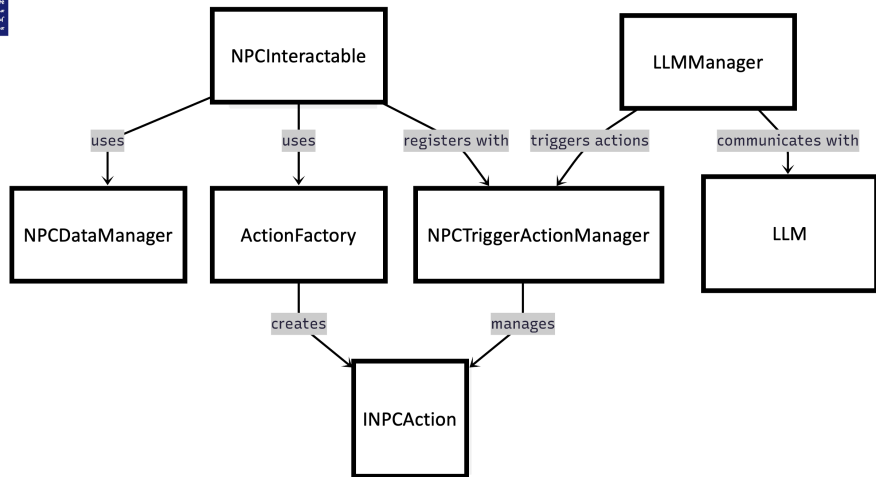
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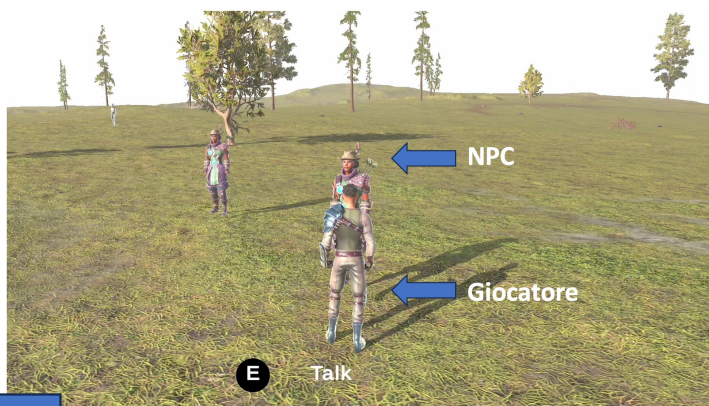
Domande?



Architettura del sistema



Obiettivo della tesi



WarnOthers Action

